



Radar Absorbing Materials – Key To Stealth

Radar absorbing materials are a special kind of polymer that convert radar energy into some other form of energy, such as heat, thus improving the stealth of a military aircraft. But there is a vast variety of such materials, each having its own characteristics and thus application

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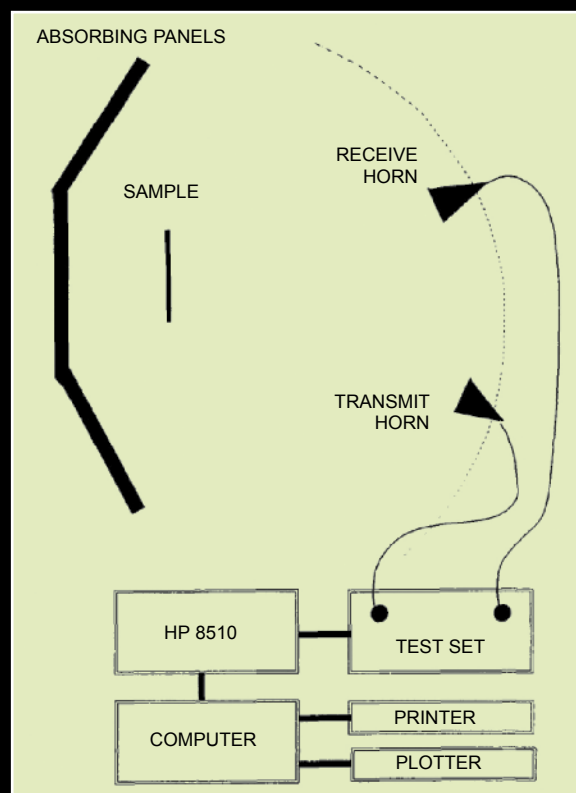
The development of radar systems before and during World War II led to research on the interaction between electromagnetic radiation at radar frequencies and various materials. One aspect of this research is finding ways to reduce the returned signal. This was done not only to reduce interference between the radar signal and surrounding buildings (for example, radar masts and support structures that could reduce the designed performance of the radar itself),

but also to help reduce hostile radar detection by properly designed anechoic systems and materials.

The Germans developed coatings for their underwater periscopes, snorkels, and command towers that achieved a nearly 26db decrease in reflection in the 112 to 195cm wavelength band. However, a number of physical deficiencies, primarily the coating's lack of survivability in harsh environments, prevented the large-scale deployment of this material.

Since World War II, there has been an increasing interest in radar absorbing materials (RAM), culminating in 'stealth' US projects for the advanced technology bomber and advanced technology fighter and the deployment of RAM on warships. These developments have been based on a synergistic approach, obtaining the reduction of the radar cross-section through several complementary methods. The basic methods used are radar absorbing structures, transparent radar structures, geometric shape modification and the use of radar absorbing materials as surface coatings. The application of these technologies to all types of defence material is of utmost importance.

The increasing use of radars in war surveillance and weapon locating, in addition to the normal or traditional function of aerial target detection, means that improved radar cloaking and



Testing various materials for RADAR absorbing properties